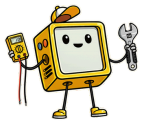
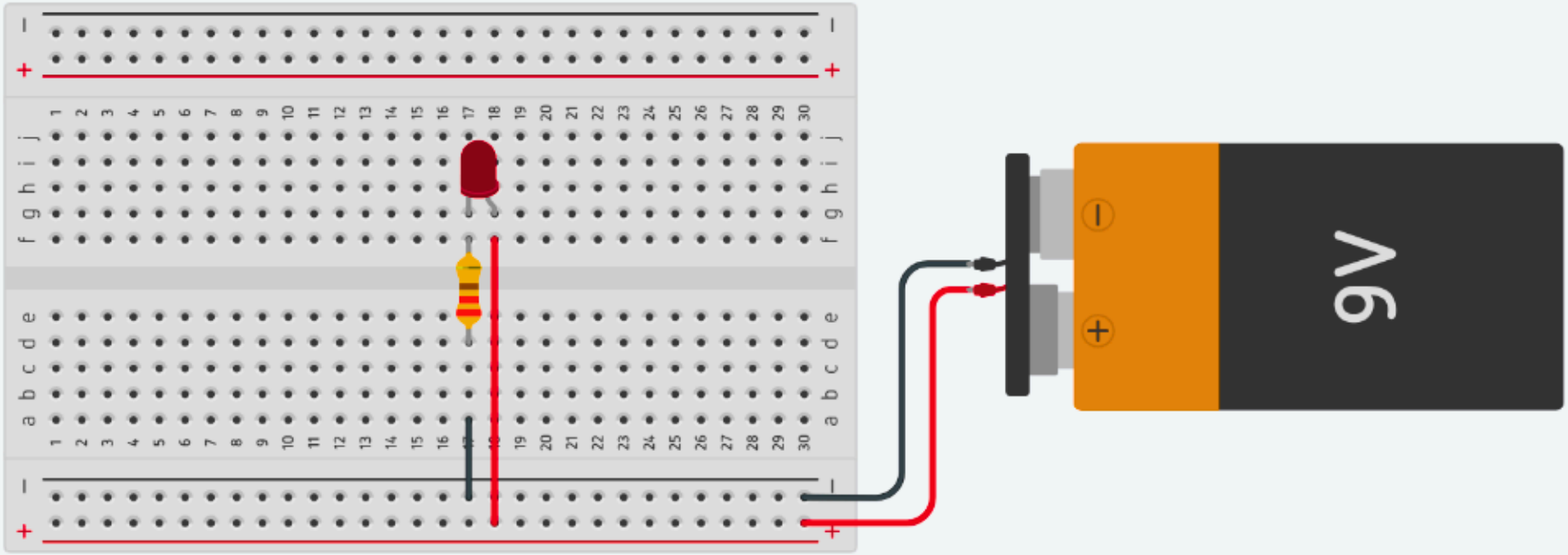
NIVEL 1



HAGASE LA LUZ



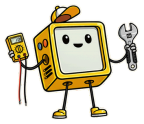
Objetivo:

La luz LED se encendera cada vez que conectes la pila

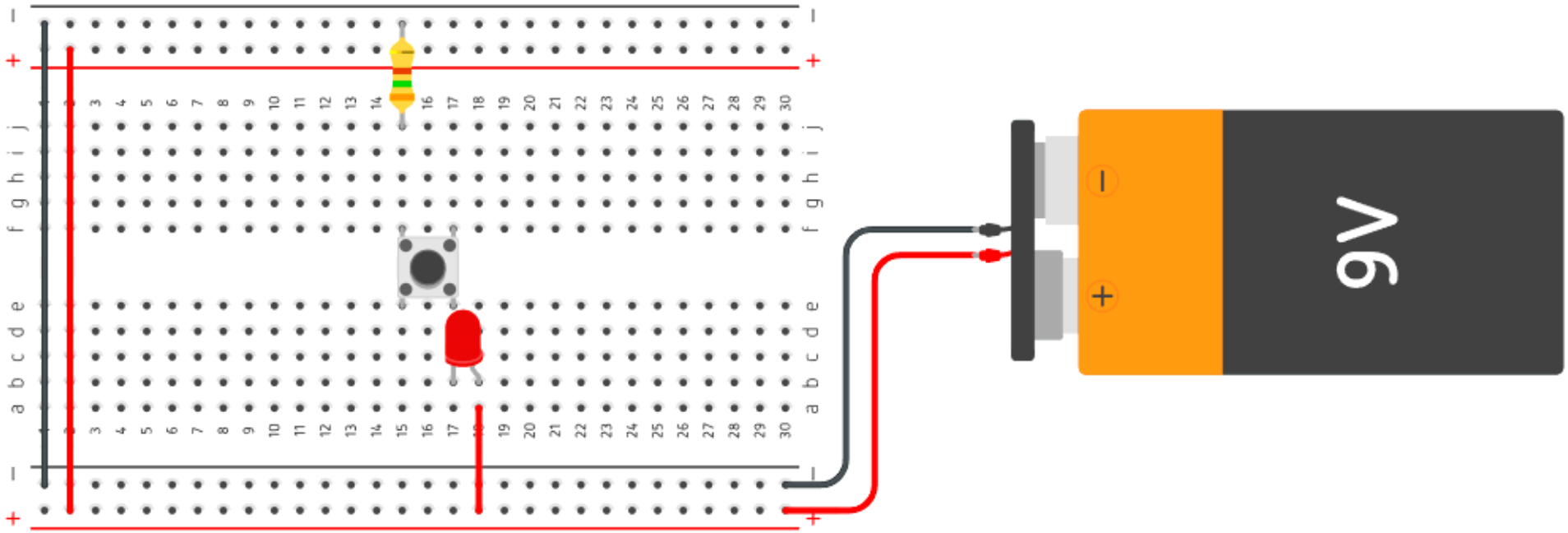
Materiales:

- 1 Led
- 1 resistor
- 1 protoboard

NIVEL 1



LED Simple



Objetivo:

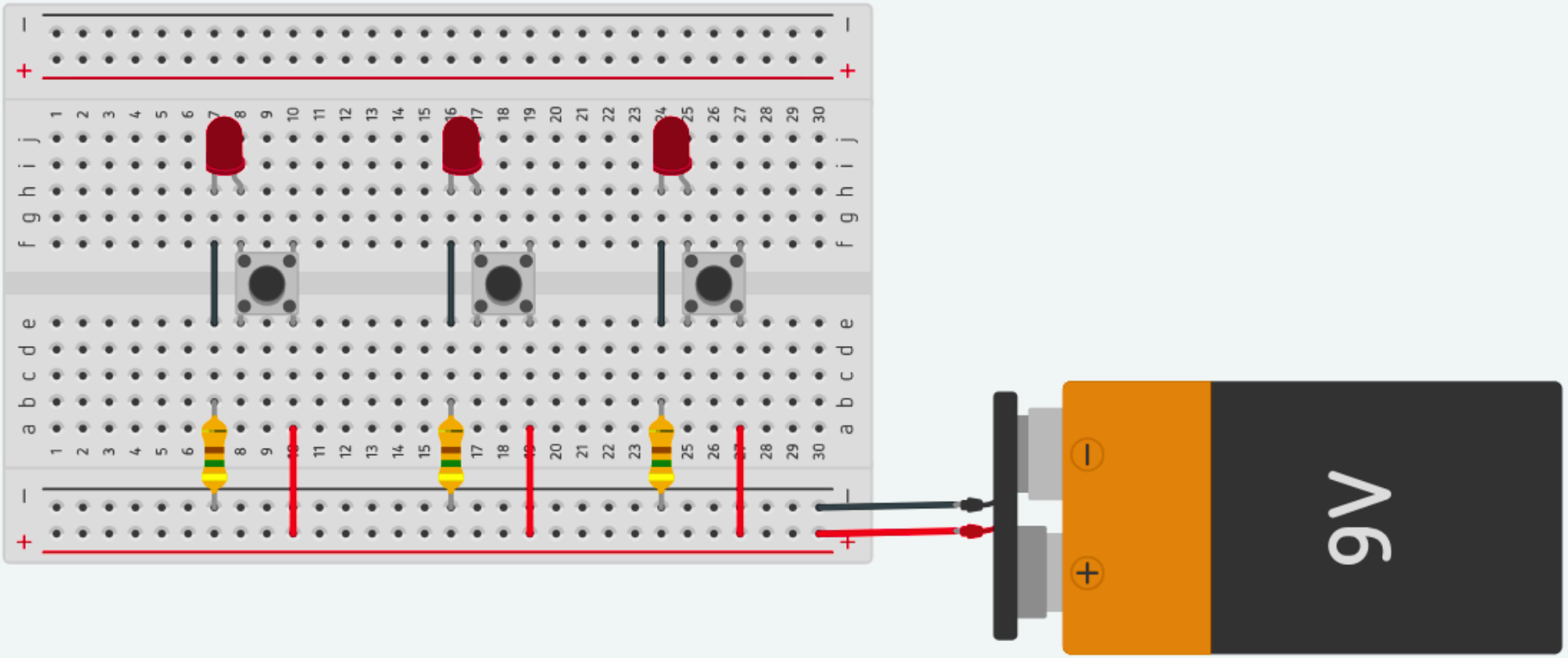
La luz LED se encendera cada vez que presiones el boton

Materiales:

- 1 Led
- 1 botón
- 1 resistor
- 1 protoboard

NIVEL 1

Piano LED



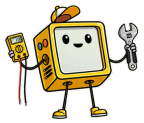
Objetivo:

Cada luz LED se encendera segun que botón este presionado

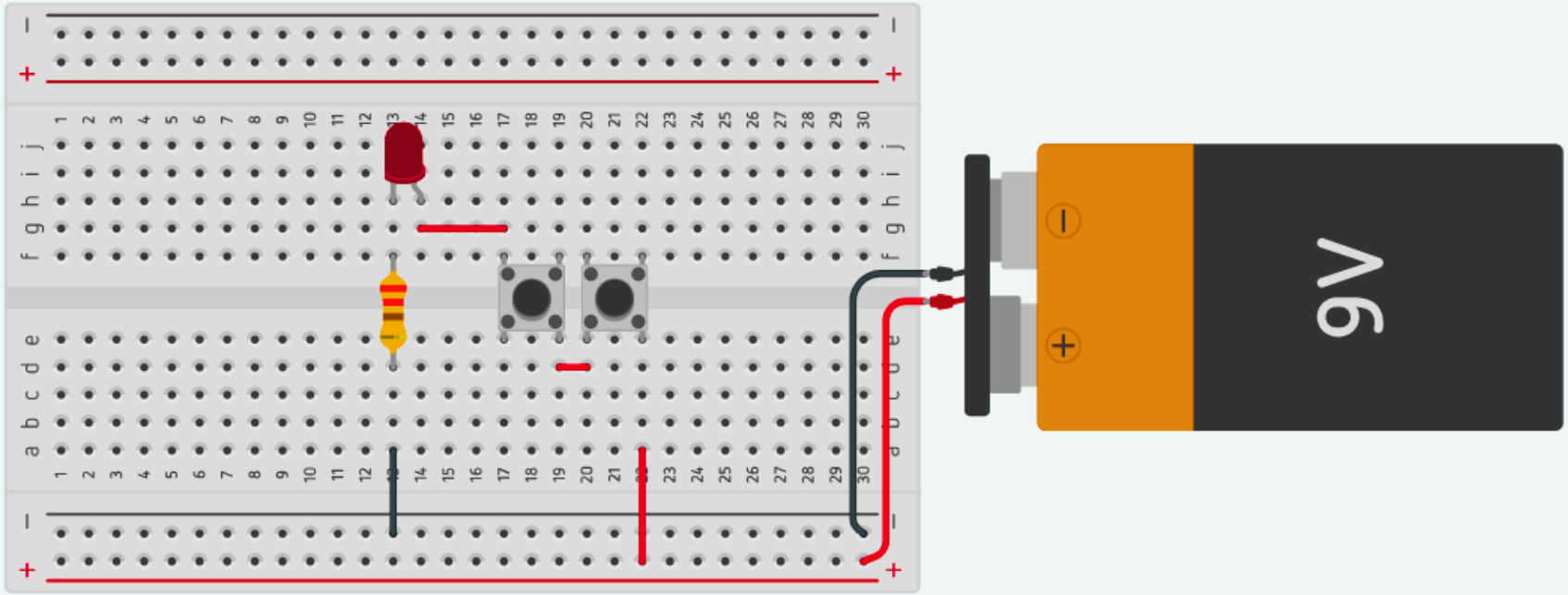
Materiales:

- 3 Leds
- 3 botones
- 3 resistores
- 1 protoboard

NIVEL 1



Compuerta AND



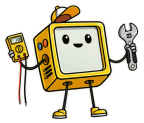
Objetivo:

Led solo se prende si ambos botones estan presionados a la vez

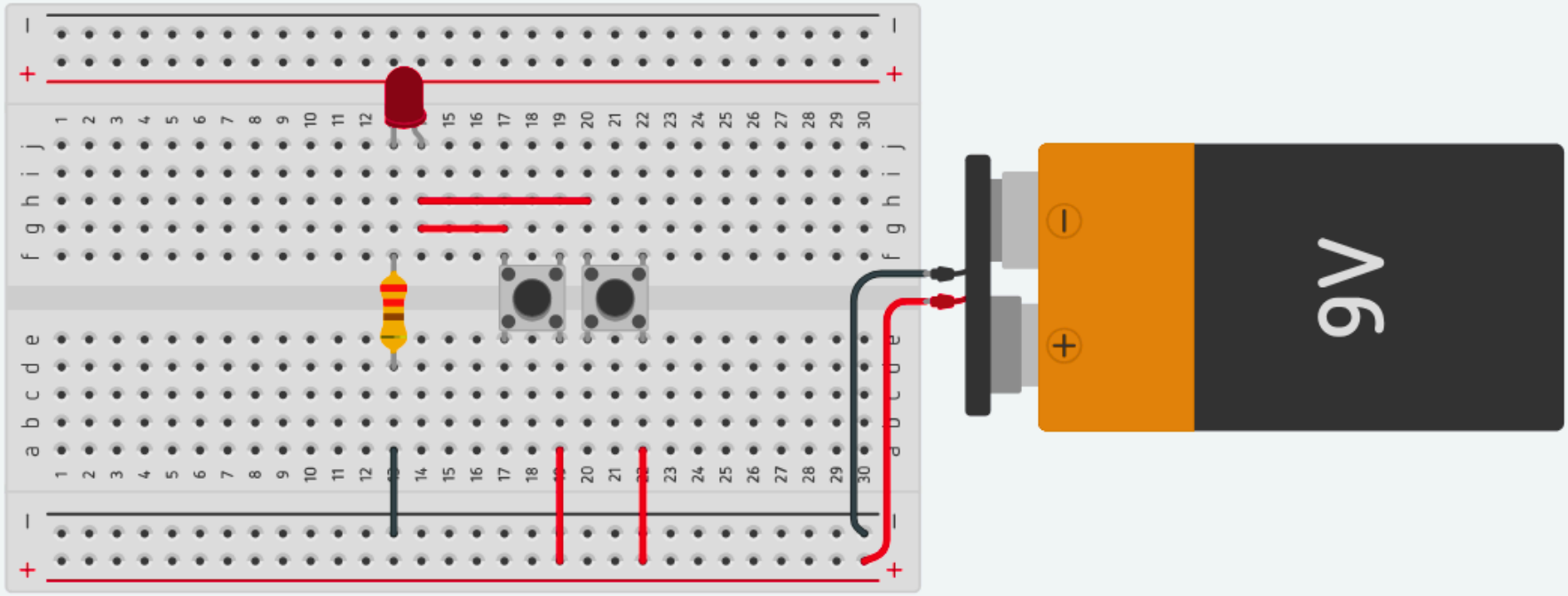
Materiales:

- 1 Led
- 2 botones
- 1 resistor
- 1 protoboard

NIVEL 1



Compuerta OR



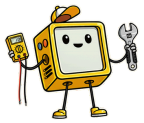
Objetivo:

Led se prende si algún boton esta presionado a la vez

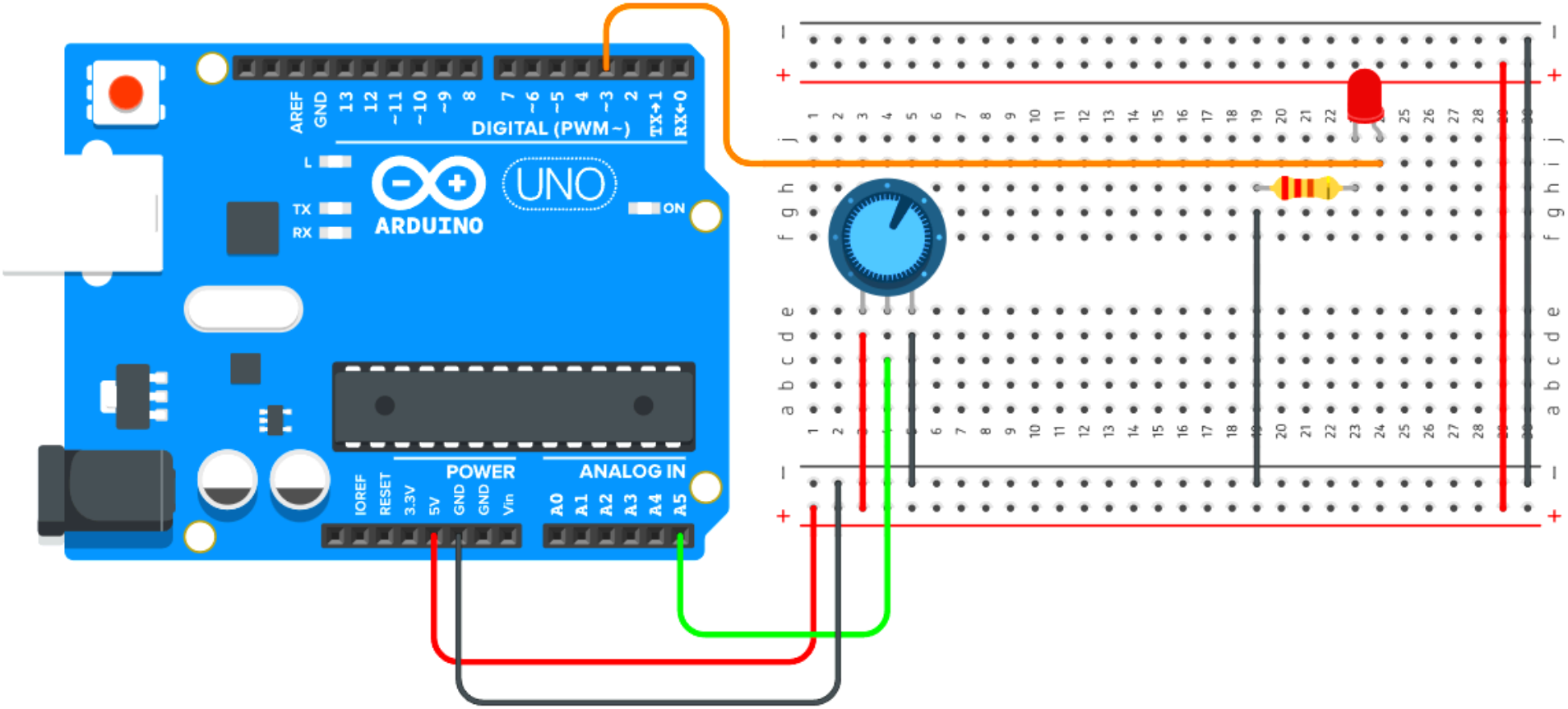
Materiales:

- 1 Led
- 2 botones
- 1 resistor
- 1 protoboard

NIVEL 1



Espantacuco



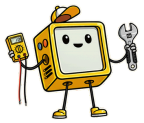
Objetivo:

Potenciometro regula intensidad de la luz

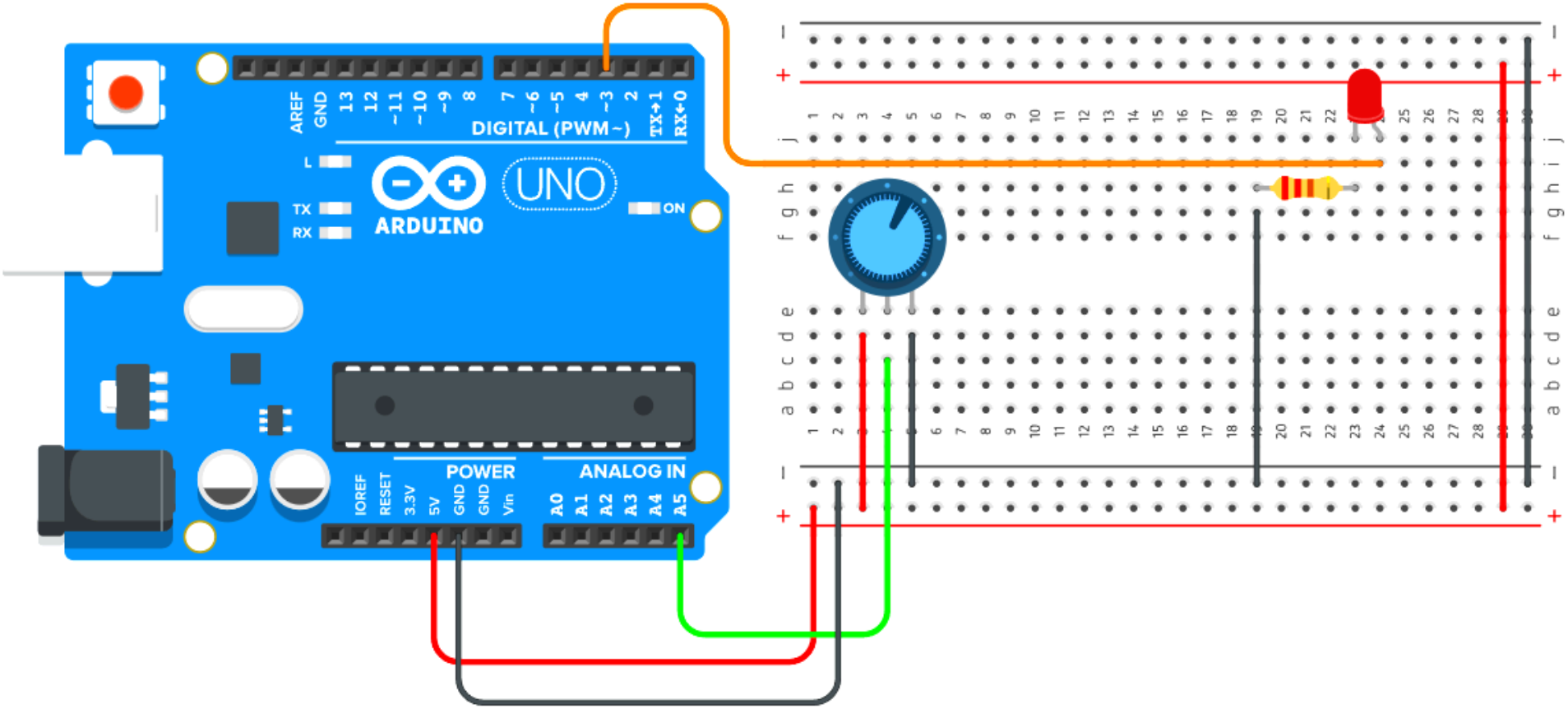
Materiales:

- 1 Led
- 1 potenciometro
- 1 resistor
- 1 protoboard
- 1 Arduino

NIVEL 1



Delay controlado



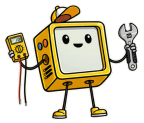
Objetivo:

El potenciometro regula el “blink rate” de la luz LED

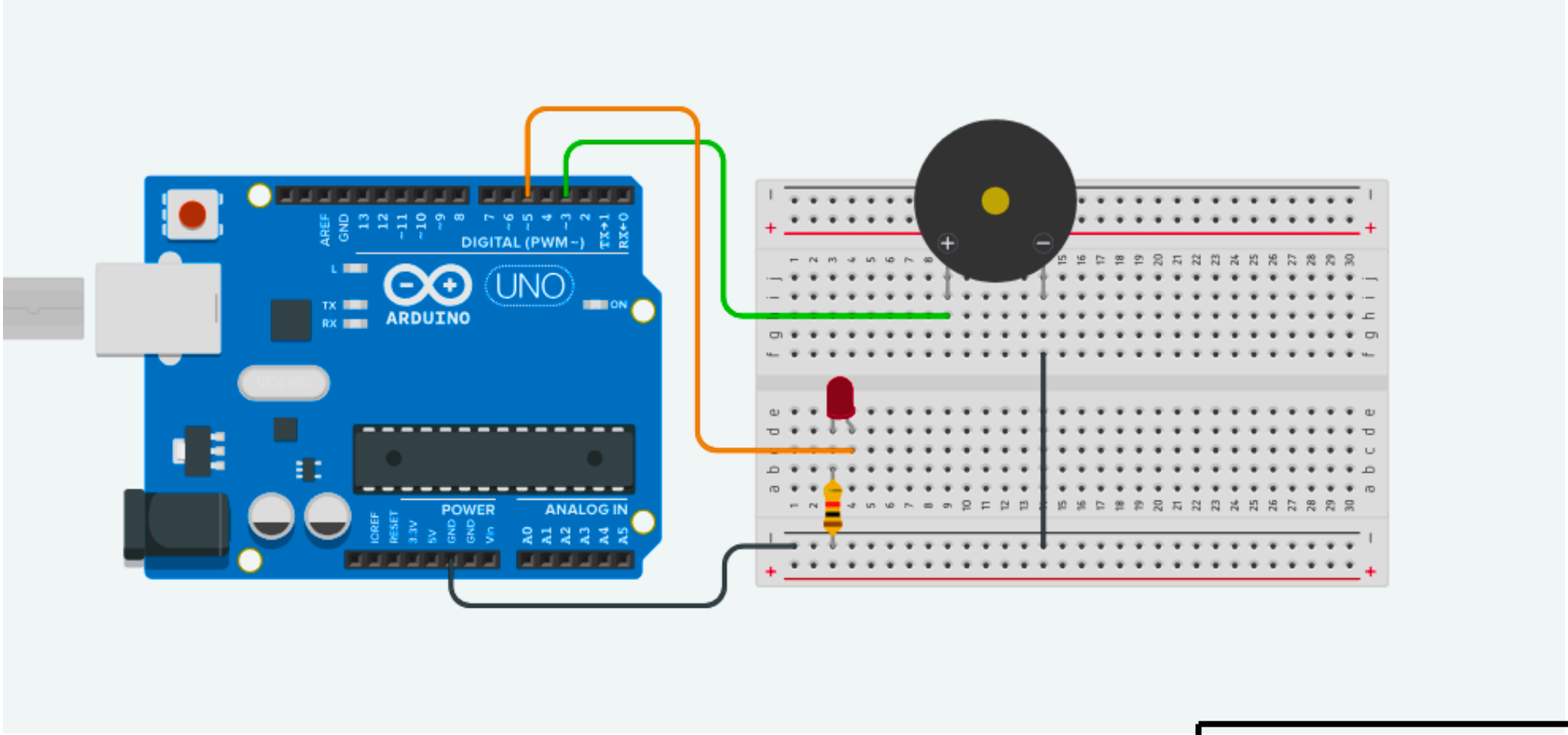
Materiales:

- 1 Led
- 1 potenciometro
- 1 resistor
- 1 protoboard
- 1 Arduino

NIVEL 1



Tiembla Mozart



Objetivo:

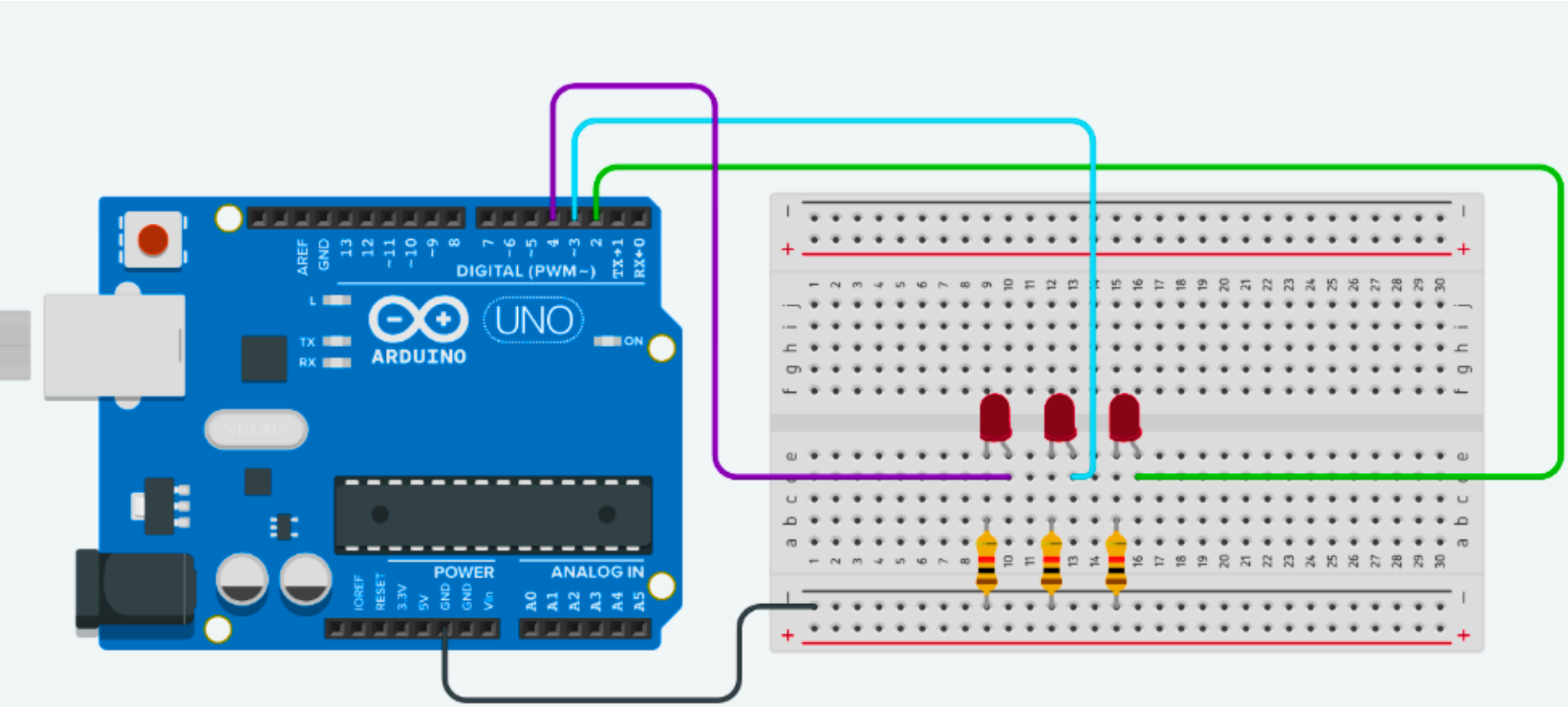
El Arduino genera musica usando el Buzzer

Materiales:

- 1 Led
- 1 Buzzer
- 1 resistor
- 1 protoboard
- 1 Arduino

NIVEL 2

Escalera autómatica



Objetivo:

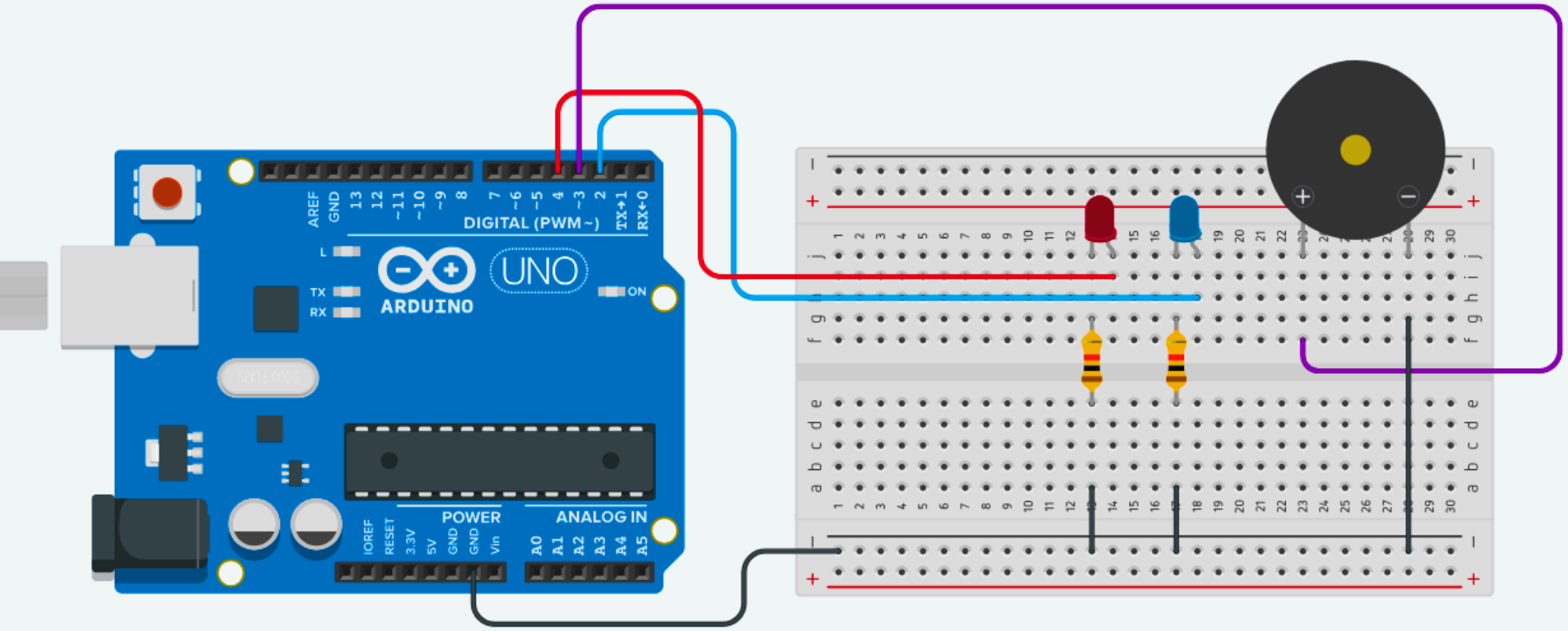
Los leds se iran intercalando simulando una escalera

Materiales:

- 3 Leds
- 3 resistores
- 1 protoboard
- 1 Arduino

NIVEL 2

Sirena Policial



Objetivo:

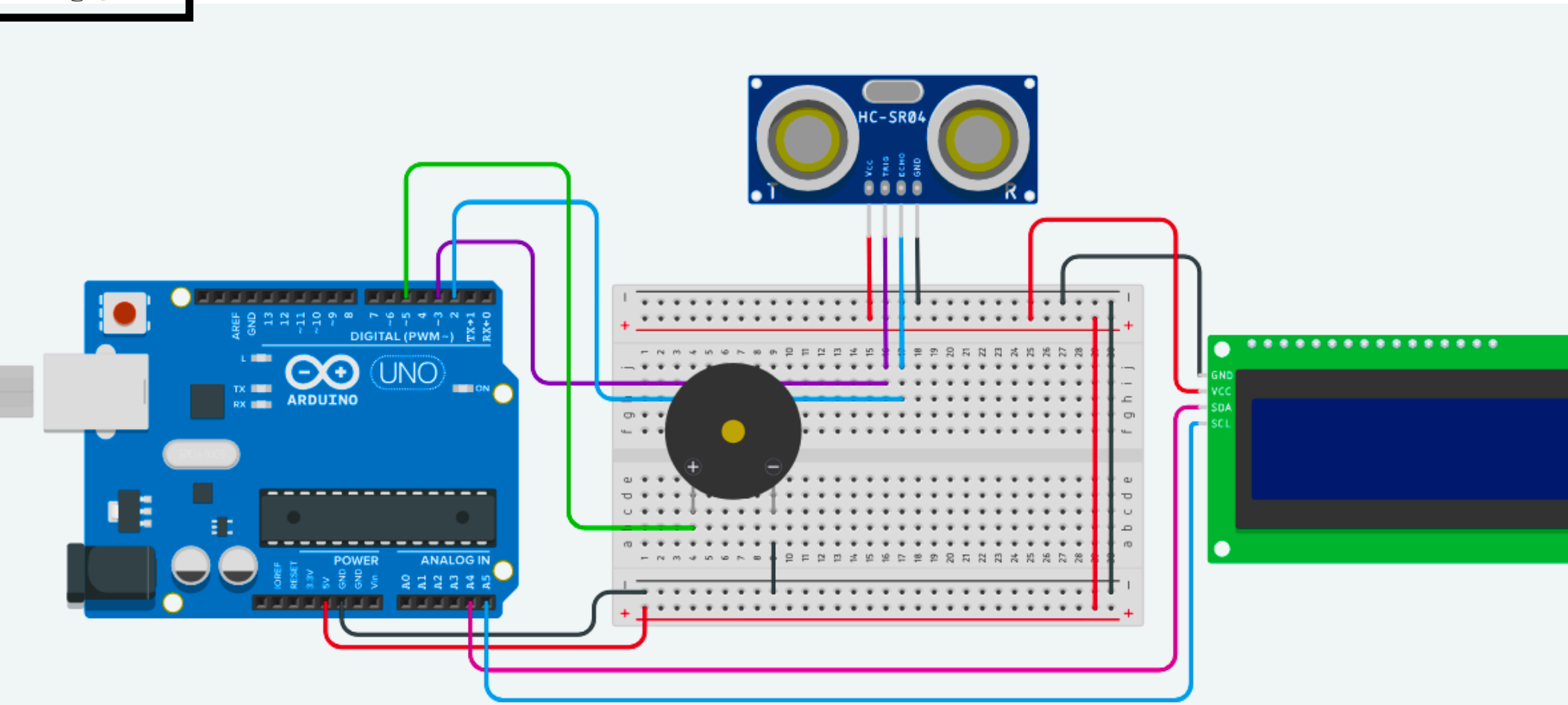
Simular una sirena policial usando 2 leds y un buzzer

Materiales:

- 2 Leds
- 1 Buzzer
- 2 resistores
- 1 protoboard
- 1 Arduino

NIVEL 2

Radar simple



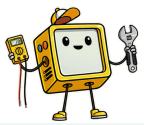
Objetivo:

El sensor ira captando distancia, el Arduino la imprimira en la pantallita LCD y el buzzer avisara segun que tan cerca hay algo

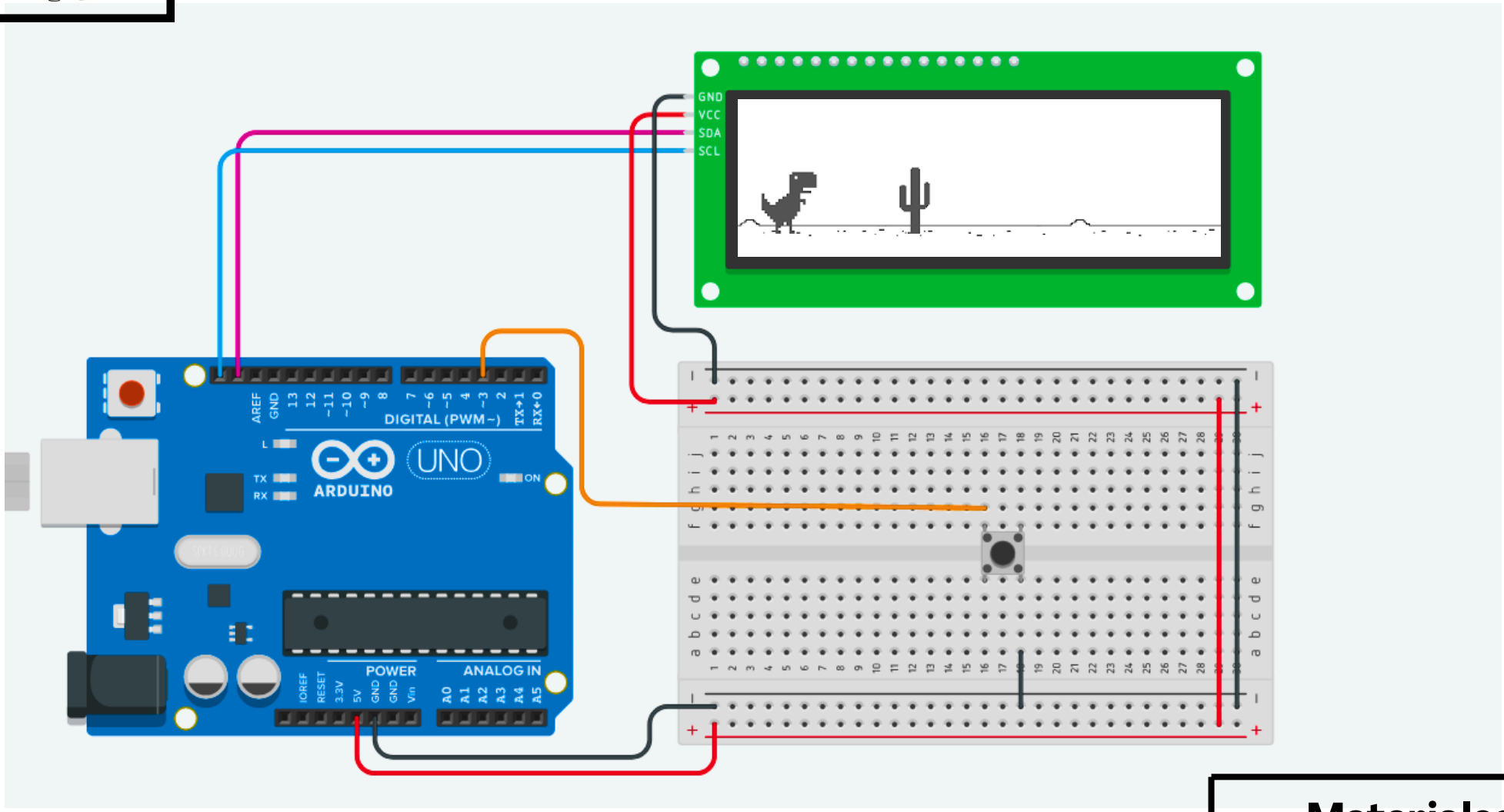
Materiales:

- 1 Sensor Ultrasónico
- 1 Buzzer
- 1 Pantalla LCD
- 1 protoboard
- 1 Arduino

NIVEL 2



DINO Game



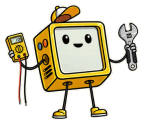
Objetivo:

La pantallita ira mostrando el clasico juego del dinosaurio de Chrome y el botón permite saltar

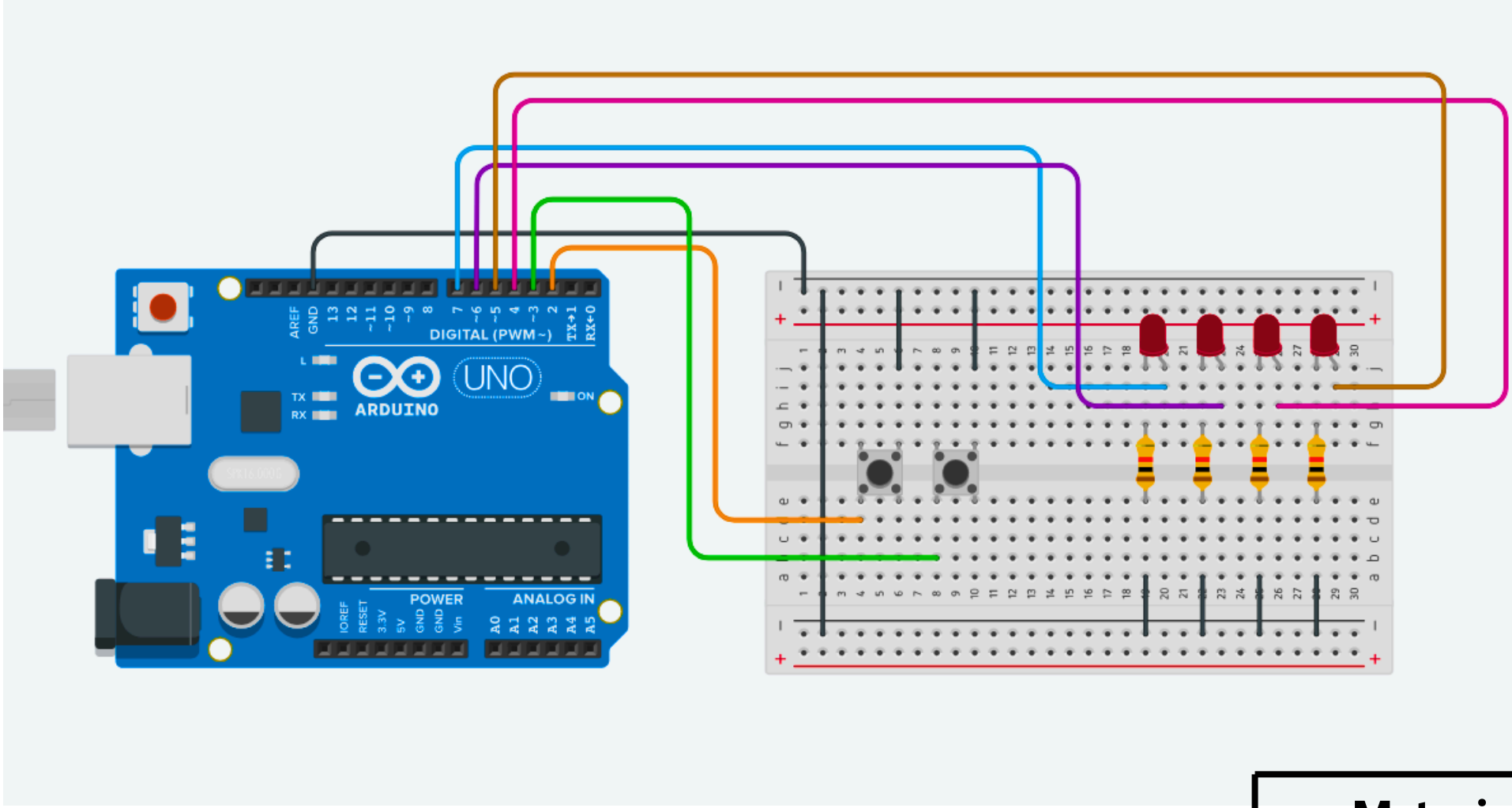
Materiales:

- 1 Pantalla LCD
- 1 Botón
- 1 protoboard
- 1 Arduino

NIVEL 3



Clave secreta



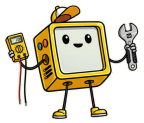
Objetivo:

Los botones hacen combinaciones raras con los LED, te toca descubrir que combinacion hace que los Leds se queden encendidos

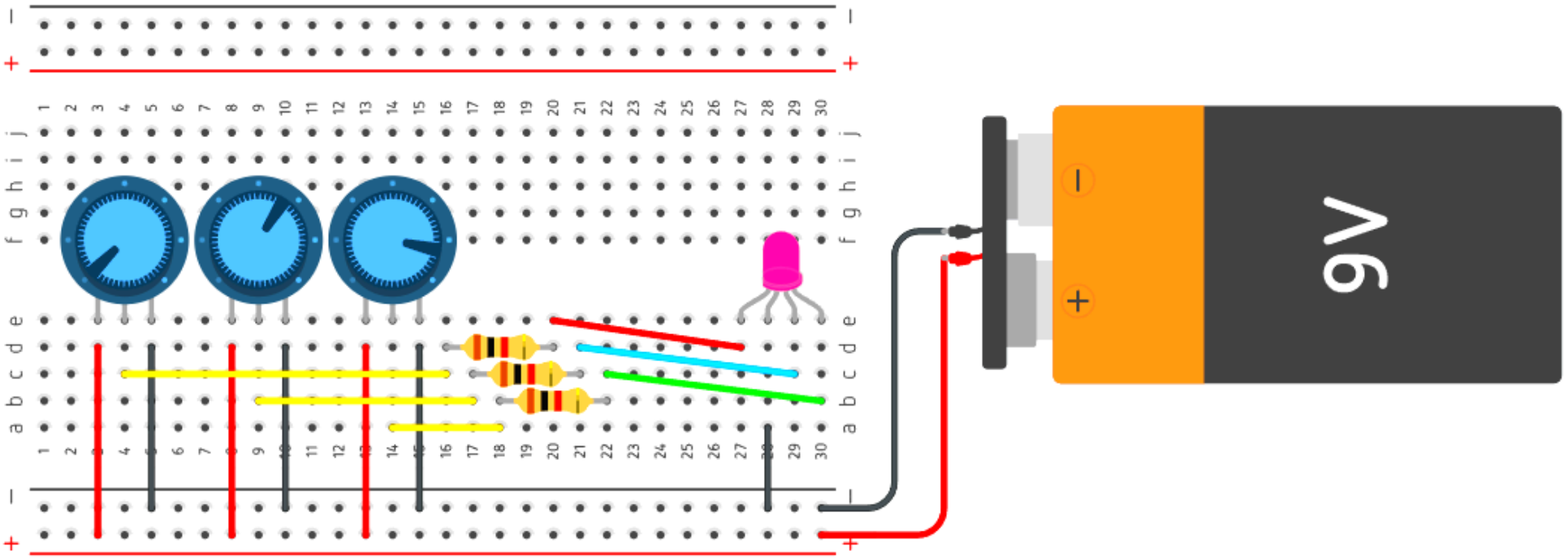
Materiales:

- 1 Arduino
- 2 Botones
- 4 Leds
- 4 resistores
- 1 protoboard

NIVEL 3



Control RGB



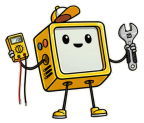
Objetivo:

La LED RGB cambiara de color segun la posicion de las perillas

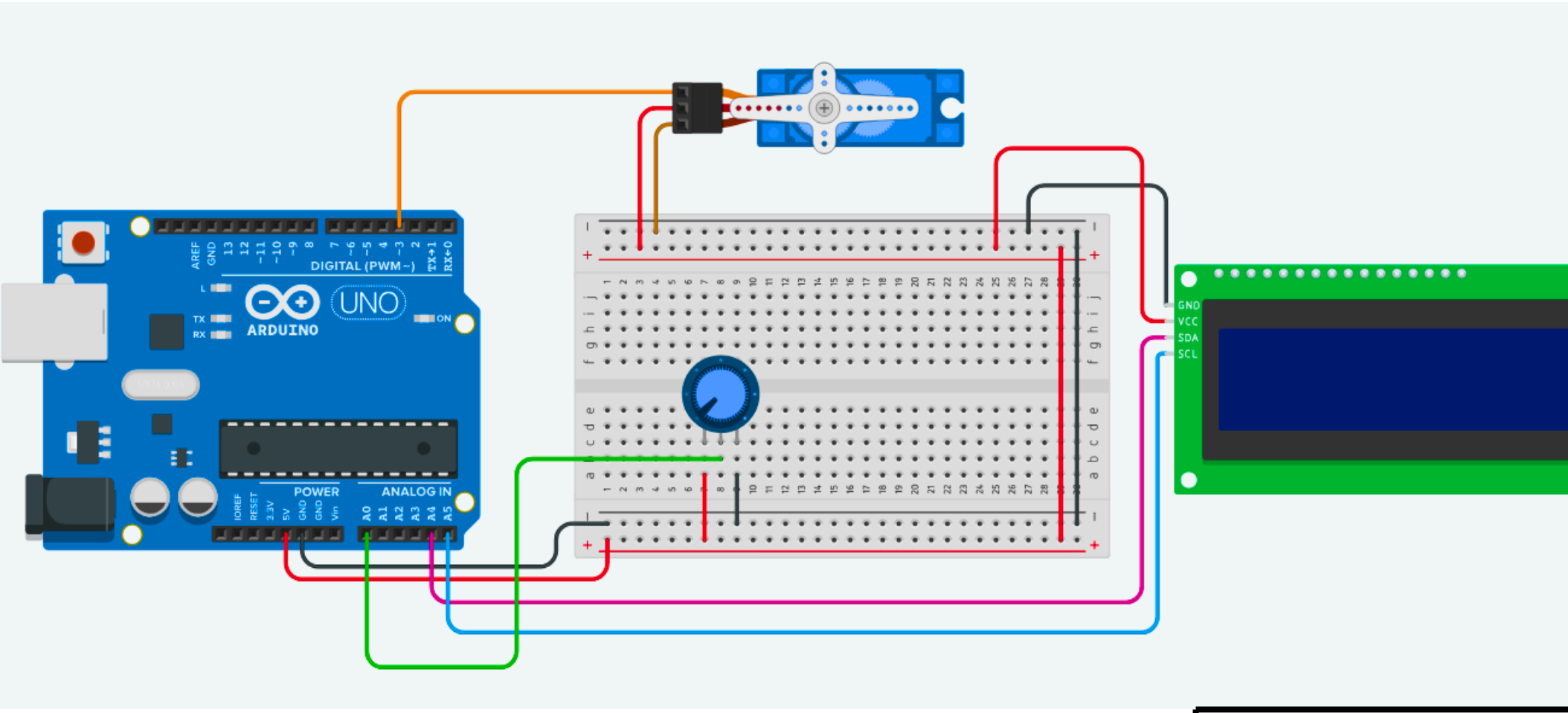
Materiales:

- 1 Led RGB
- 3 Potenciometros
- 3 resistores
- 1 protoboard

NIVEL 3



Servo loco



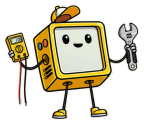
Objetivo:

El servo cambiará de posición segun el potenciometro y la pantalla LCD mostrará su ángulo actual

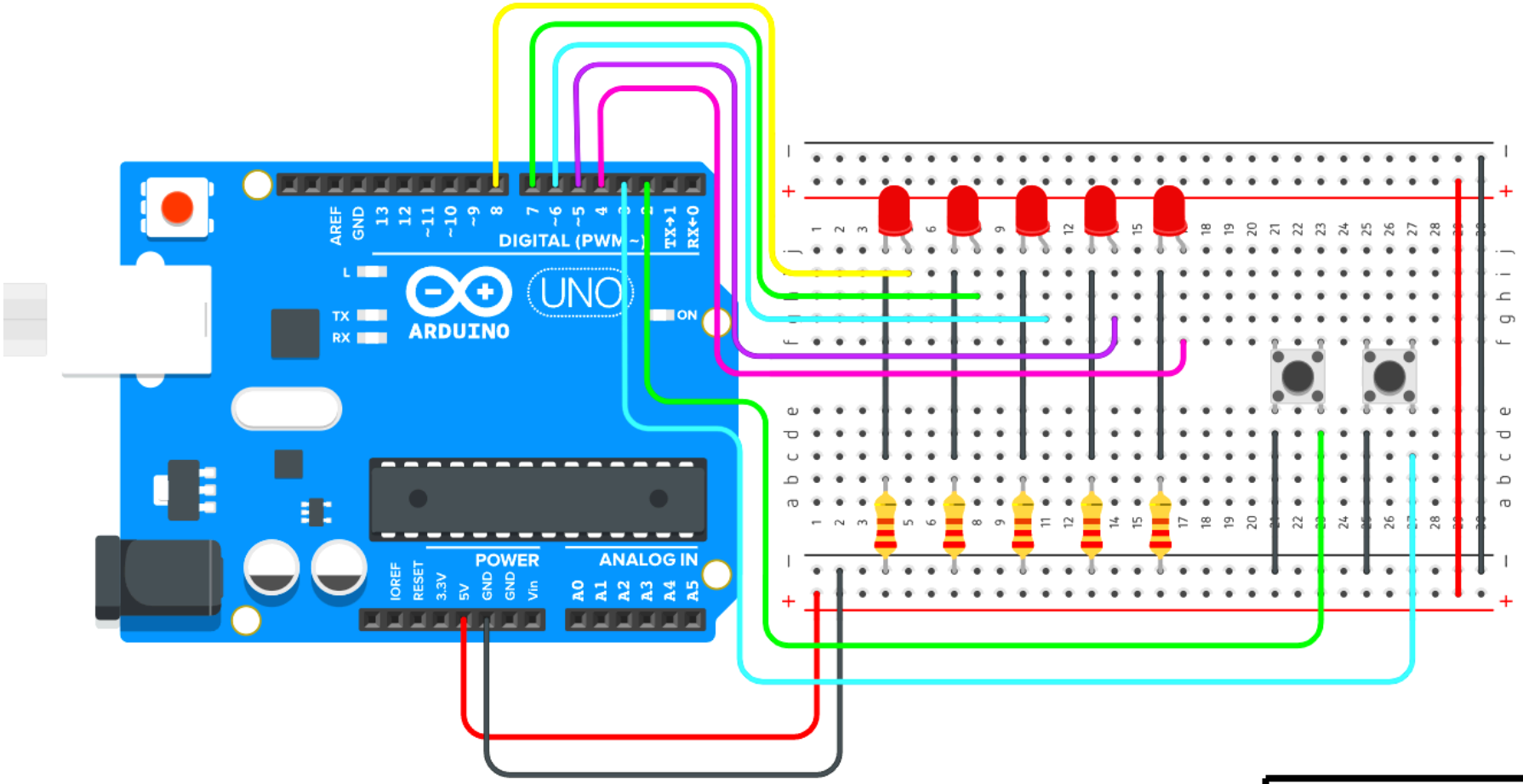
Materiales:

- 1 Arduino
- 1 Servo
- 1 Pantalla LCD
- 1 Potenciometro
- 1 protoboard

NIVEL 3



LED Game



Objetivo:

Los botones van haciendo que cambien los LED encendidos

Materiales:

- 1 Arduino
- 5 Leds
- 5 resistores
- 2 botones
- 1 protoboard